

No almost complete embedding for the original Kalton operator-space pair of arXiv:2106.15022

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Status. Claimed full solution, likely valid, pending human review. The proof also gives a stronger obstruction for arbitrary ℓ_1 -sums of minimal operator spaces.

1 Source question

Braga's Theorem 1.8 constructs operator spaces X and Y such that X embeds linearly isomorphically, but not completely isomorphically, into Y , while the bounded subsets of X and Y are almost completely coarsely equivalent [1]. The paragraph immediately following the theorem asks whether the *specific* spaces constructed there also satisfy that X does not almost completely isomorphically embed into Y .

Theorem 1.8. *There are operator spaces X and Y so that*

- (1) *X linearly isomorphically embeds into Y ,*
- (2) *X does not completely isomorphically embed into Y , and*
- (3) *the bounded subsets of X and Y are almost completely coarsely equivalent.*

We point out that we do not know if the examples of X and Y provided by Theorem 1.8 are such that X does not *almost* completely isomorphically embed into Y . Also, the examples in Theorem 1.8 are nonseparable.

Figure 1: Theorem 1.8 and the source question on PDF page 4 of [1].

For the explicit construction, put $H = R \cap C$. Let $Q : L_\infty[0, 1] \rightarrow H$ be the complete quotient adjoint to the standard complete embedding $R + C \rightarrow L_1[0, 1]$. On the vector space $L_\infty[0, 1]$, let

$$\|y\|_{Y_j} = \max\{2^{-j}\|y\|_\infty, \|Qy\|_H\}, \quad \mathcal{Z}(Q) = \left(\bigoplus_{j \geq 1} Y_j\right)_{\ell_1}.$$

The source pair is

$$Y = \mathcal{Z}(Q), \quad X = H \oplus \ker \tilde{Q}.$$

An almost complete isomorphic embedding of an operator space E into F means a sequence $T_n : E \rightarrow F$ and constants $a, b > 0$ such that

$$a\|z\|_{M_n(E)} \leq \|(T_n)_n z\|_{M_n(F)} \leq b\|z\|_{M_n(E)} \tag{1}$$

for every n and $z \in M_n(E)$.

2 Proof intuition

There are two steps. First, the quotient coordinates of every bounded map $H \rightarrow \mathcal{Z}(Q)$ form a compact operator into $\ell_1(H)$. Therefore, on a suitable infinite-dimensional Hilbert subspace, those coordinates are as small as desired. This reduces a hypothetical almost complete embedding into $\mathcal{Z}(Q)$ to one into

$$W = \left(\bigoplus_{j \geq 1} \text{MIN}(L_\infty[0, 1]) \right)_{\ell_1}.$$

Second, fix one map $T_n : H \rightarrow W$. Weak compactness gives a finite coordinate head whose scalar-norm tail is tiny. On that finite head, concentration on high-dimensional Euclidean spheres supplies an n -dimensional Hilbert subspace F on which all coordinate norms are nearly constant simultaneously. Their operator norms on F then have bounded sum. For an orthonormal basis $x_1, \dots, x_n$ of F , the row matrix $z = \sum_k e_{1k} \otimes x_k$ has norm $\sqrt{n}$ in $R \cap C$, whereas its image in the finite minimal head has uniformly bounded norm. The tiny tail stays bounded after amplification. This contradicts the lower estimate in (1).

3 Three auxiliary lemmas

Lemma 1 (Uniform tails). *Let (E_j) be Banach spaces and $Z = (\bigoplus_j E_j)_{\ell_1}$. Every relatively weakly compact subset $K \subset Z$ has uniformly small coordinate tails. Consequently, if G is reflexive and $T : G \rightarrow Z$ is bounded, then*

$$\|(I - P_N)T\| \longrightarrow 0,$$

where P_N is the projection onto the first N coordinates.

Proof. Suppose uniform tails fail. A gliding-hump selection gives $\varepsilon > 0$, pairwise successive finite intervals $I_m \subset \mathbb{N}$, and $z^{(m)} \in K$ such that

$$\sum_{j \in I_m} \|z_j^{(m)}\| > \varepsilon.$$

Choose $f_{m,j} \in B_{E_j^*}$, adjusting complex phases if necessary, so that

$$\left| \sum_{j \in I_m} f_{m,j}(z_j^{(m)}) \right| > \varepsilon.$$

The map

$$A : Z \rightarrow \ell_1, \quad (Az)_m = \sum_{j \in I_m} f_{m,j}(z_j),$$

is contractive because the intervals are disjoint. Thus $A(K)$ is relatively weakly compact. By the Schur property of ℓ_1 , it is relatively norm compact and hence has uniformly small scalar coordinate tails. This contradicts $|(Az^{(m)})_m| > \varepsilon$. The last assertion follows because $T(B_G)$ is relatively weakly compact. $\square$

Lemma 2 (Simultaneous Hilbertian stabilization). *Let G be an infinite-dimensional Hilbert space and let $p_1, \dots, p_N$ be continuous seminorms on G satisfying*

$$\sum_{j=1}^N p_j(x) \leq C \|x\|_2 \quad (x \in G). \quad (2)$$

For every $d \in \mathbb{N}$ and $\varepsilon > 0$, there is a d -dimensional subspace $F \subset G$ such that

$$\sum_{j=1}^N \sup_{x \in S_F} p_j(x) \leq C + \varepsilon. \quad (3)$$

Proof. We first recall the following additive form of Euclidean-sphere stabilization. Given d, L and $\delta > 0$, there is D such that, whenever E is a Euclidean space of dimension at least D and p is a seminorm with $p(x) \leq L\|x\|_2$, some d -dimensional $F \subset E$ satisfies

$$\sup_{S_F} p - \inf_{S_F} p < \delta. \quad (4)$$

Indeed, p is L -Lipschitz on S_E . Lévy's concentration lemma says that it is exponentially concentrated about a median. For a random isometric copy of ℓ_2^d in E , take a ρ -net of S^{d-1} of cardinality at most $(1 + 2/\rho)^d$. If $L > 0$, set $t = \delta/4$ and $\rho = \delta/(4L)$ (decreasing ρ to at most 1 causes no problem). Lévy's estimate and a union bound show, once $\dim E$ is large enough in terms of d, L, δ , that every net value differs from the median by at most t . Every sphere point then differs from the median by at most $t + L\rho \leq \delta/2$, so the oscillation is at most δ . The case $L = 0$ is immediate. This is the standard concentration proof of the additive Dvoretzky lemma [5, Chapter 2].

Choose positive δ_j with $\sum_j \delta_j < \varepsilon$. Starting with a sufficiently large finite-dimensional subspace of G , apply (4) successively to $p_1, \dots, p_N$, choosing the intermediate dimensions backwards so that the final dimension is d . Previously obtained oscillation estimates persist on further subspaces. On the resulting F , put $a_j = \inf_{x \in S_F} p_j(x)$. Then

$$\sup_{S_F} p_j \leq a_j + \delta_j.$$

For any fixed $x \in S_F$, (2) gives $\sum_j a_j \leq \sum_j p_j(x) \leq C$. Summing the preceding inequalities proves (3). $\square$

Lemma 3 (The minimal row identity). *Let E be a Banach space, $S : F \rightarrow E$ a bounded map on an n -dimensional Hilbert space, and $x_1, \dots, x_n$ an orthonormal basis of F . Then*

$$\left\| \sum_{k=1}^n e_{1k} \otimes Sx_k \right\|_{M_n(\text{MIN}(E))} = \|S\|. \quad (5)$$

For the same orthonormal family regarded in $H = R \cap C$,

$$\left\| \sum_{k=1}^n e_{1k} \otimes x_k \right\|_{M_n(H)} = \sqrt{n}. \quad (6)$$

Proof. By the definition of the minimal operator-space structure, the left side of (5) is

$$\sup_{\phi \in B_{E^*}} \left(\sum_{k=1}^n |\phi(Sx_k)|^2 \right)^{1/2} = \|(S|_F)^*\| = \|S\|.$$

For (6), unitary homogeneity lets us take x_k to be the first n standard basis vectors of ℓ_2 . The row component of the displayed matrix has norm $\sqrt{n}$, while its column component has norm 1. The $R \cap C$ norm is their maximum. $\square$

4 The minimal-sum obstruction

Theorem 4. *For every sequence of Banach spaces (E_j) , the operator space*

$$W_E = \left(\bigoplus_{j \geq 1} \text{MIN}(E_j) \right)_{\ell_1}$$

does not admit an almost complete isomorphic embedding of $R \cap C$.

Proof. Suppose that maps $T_n : H = R \cap C \rightarrow W_E$ satisfy (1) for fixed $a, b > 0$. Fix $\eta > 0$ and then a large n . At scalar level, $\|T_n\| \leq b$. Since H is reflexive, Lemma 1 gives N such that

$$\|(I - P_N)T_n\| \leq \eta n^{-3/2}. \quad (7)$$

Write $S_j : H \rightarrow E_j$ for the j th coordinate of T_n . The underlying Banach norm of W_E is the ℓ_1 norm, so

$$\sum_{j=1}^N \|S_j x\| \leq b \|x\|_2.$$

Apply Lemma 2 to $p_j(x) = \|S_j x\|$. We obtain an n -dimensional $F \subset H$ such that

$$\sum_{j=1}^N \|S_j|_F\| \leq b + \eta. \quad (8)$$

Let $x_1, \dots, x_n$ be an orthonormal basis of F and set

$$z_F = \sum_{k=1}^n e_{1k} \otimes x_k \in M_n(H).$$

By Lemma 3, $\|z_F\| = \sqrt{n}$. The complete isometry of each coordinate inclusion into the operator-space ℓ_1 sum, the triangle inequality, (5), and (8) give

$$\|(P_N T_n)_n z_F\| \leq \sum_{j=1}^N \|(S_j)_n z_F\| = \sum_{j=1}^N \|S_j|_F\| \leq b + \eta. \quad (9)$$

For every bounded map A between operator spaces, $\|A_n\| \leq n\|A\|$. (Represent the target concretely and use that the norm of an n by n operator matrix is at most n times its largest entry norm.) Hence (7) and (6) imply

$$\|((I - P_N)T_n)_n z_F\| \leq n \eta n^{-3/2} \sqrt{n} = \eta. \quad (10)$$

Combining (9) and (10),

$$\|(T_n)_n z_F\| \leq b + 2\eta.$$

But the lower estimate in (1) gives $\|(T_n)_n z_F\| \geq a\sqrt{n}$, a contradiction for large n . $\square$

5 Compact quotient coordinates and the original pair

Set

$$W = \left(\bigoplus_{j \geq 1} \text{MIN}(L_\infty[0, 1]) \right)_{\ell_1}, \quad U = \left(\bigoplus_{j \geq 1} H \right)_{\ell_1}.$$

Here the canonical operator-space structure of the commutative C^* -algebra $L_\infty[0, 1]$ is its minimal operator-space structure. For $y = (y_j) \in \mathcal{Z}(Q)$, define

$$P_1 y = (2^{-j} y_j)_j \in W, \quad P_2 y = (Q y_j)_j \in U.$$

The map $\Gamma = (P_1, P_2) : \mathcal{Z}(Q) \rightarrow W \oplus_\infty U$ is a complete isomorphic embedding with

$$\|\Gamma\|_{\text{cb}} \leq 1, \quad \|\Gamma^{-1} : \Gamma(\mathcal{Z}(Q)) \rightarrow \mathcal{Z}(Q)\|_{\text{cb}} \leq 2. \quad (11)$$

Indeed, coordinatewise this is the defining maximum norm of Y_j . Regrouping the two ℓ_1 sums is completely contractive; the reverse map is the sum of its two coordinate maps, each completely contractive, and so has cb-norm at most 2.

Lemma 5. *For every bounded $u : H \rightarrow \mathcal{Z}(Q)$, the quotient-coordinate operator*

$$V_u : H \rightarrow U, \quad V_u x = (Q u_j x)_j = P_2 u(x),$$

is compact.

Proof. First consider a bounded $T : H \rightarrow L_\infty[0, 1]$. We claim that $QT : H \rightarrow H$ is compact. Otherwise, since its domain and range are Hilbert spaces, there is an infinite-dimensional closed $E \subset H$ on which $A = QT$ is bounded below. The range $A(E)$ is closed and

$$\Pi = T|_E (A|_E)^{-1} P_{A(E)} Q : L_\infty[0, 1] \rightarrow T(E)$$

is a bounded projection onto a copy of ℓ_2 . This contradicts the classical complemented-subspace theorem for L_∞ used in Braga's proof of Proposition 4.6 [1, 4]. Thus each coordinate $Q u_j$ is compact.

The map $V_u : H \rightarrow \ell_1(H)$ is bounded by the definition of $\mathcal{Z}(Q)$. Since H is reflexive, it is weakly compact. Lemma 1 shows that its coordinate tails converge to zero in operator norm. Its finite coordinate heads are compact, so V_u is a norm limit of compact operators. $\square$

Theorem 6. *The operator space $H = R \cap C$ does not almost completely isomorphically embed into the original $\mathcal{Z}(Q)$ of Theorem 1.8 in [1].*

Proof. Suppose maps $u^{(n)} : H \rightarrow \mathcal{Z}(Q)$ satisfy (1). Fix n . By Lemma 5, $P_2 u^{(n)}$ is compact. Approximate it in operator norm by a finite-rank map and pass to the infinite-dimensional kernel of that approximant. We obtain an infinite-dimensional $F_n \subset H$ on which

$$\|P_2 u^{(n)}|_{F_n}\| < \frac{a}{4n}.$$

Every infinite-dimensional Hilbert subspace of $R \cap C$ is the range of a complete isometry $J_n : H \rightarrow F_n$, because row and column norms are invariant under Hilbert-space unitaries.

Put $T_n = P_1 u^{(n)} J_n : H \rightarrow W$. From (1) and (11),

$$\|\Gamma_n u^{(n)} J_n z\| \geq \frac{a}{2} \|z\|.$$

The P_2 component has n th amplification norm less than $a/4$. Therefore

$$\|(T_n)_nz\| \geq \frac{a}{2}\|z\|, \quad \|(T_n)_nz\| \leq b\|z\|$$

for all $z \in M_n(H)$. Thus (T_n) would almost completely embed H into W , contradicting Theorem 4. $\square$

Corollary 7 (Answer to the source question). *For the exact operator spaces*

$$X = (R \cap C) \oplus \ker \tilde{Q}, \quad Y = \mathcal{Z}(Q)$$

constructed in Theorem 1.8 of [1], X does not almost completely isomorphically embed into Y .

Proof. The canonical $R \cap C$ summand of X is completely isometric. Restricting any hypothetical almost complete embedding $X \rightarrow Y$ to this summand would contradict Theorem 6. $\square$

6 Verification notes and novelty scope

The proof is qualitative; no computation is used. The main points checked independently in the proof are:

- Lemma 1 is proved for arbitrary Banach-valued ℓ_1 sums, so no hidden reflexivity of the coordinate spaces is assumed.
- Lemma 2 uses only finite families. Intermediate dimensions may be enormous, but the domain Hilbert space is infinite dimensional and the final required dimension is exactly n .
- The equality in Lemma 3 follows directly from the definition of $\text{MIN}(E)$; it does not rely on a special representation of L_∞ .
- The tail exponent $n^{-3/2}$ exactly compensates the general loss $\|A_n\| \leq n\|A\|$ and the input row norm $\sqrt{n}$.
- The quotient-coordinate reduction uses only compactness and the complete distortion bound 2 in (11); all constants have strict slack.

The required bounded novelty search was performed on 19 July 2026. The four local run indexes and the locally stored arXiv sources 2106.15022 and 2211.11854 were searched for the arXiv id, the exact question phrase, “ $R \cap C$ ”, “almost complete isomorphic embedding”, “minimal”, and “ ℓ_1 -sum”. Bounded external searches used the same phrases, the title, and the authors Braga and Oikhberg, restricted in the technical search to primary arXiv records. The search found the source paper, Braga–Oikhberg’s different-space existential strengthening [2], and the later small-scale nonlinear paper, but no proof for the exact original $L_\infty \rightarrow R \cap C$ pair and no statement of Theorem 4. This is evidence only, not a guarantee of novelty.

Braga–Oikhberg say that their existential theorem addresses the motivation of the question after Theorem 1.8, but their construction uses the different quotient $\text{MAX}(L_1) \rightarrow \text{MIN}(\ell_2)$ and different spaces. It does not settle the exact-space assertion in Corollary 7.

Human-review recommendation. Review first the simultaneous stabilization lemma and the minimal row identity, then the factor-2 regrouping estimate (11). If these three points are accepted, the remaining argument is a direct compactness-and-amplification contradiction.

References

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